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Chapter-1,

1. What is data visualaization ?

Data visualaization is the graphical representation of data using charts graphs and diagrams to understand information easily.

eg: Bar chart

2. State two adv and dis adv of DV

Adv	Dis-adv
* helps identify trends quickly	* Can be misleading if wrong chart is used
* Makes complex data easy to understand.	* Over simplifies Complex data

3. How DV usefull in Big data ?

DV helps and analyze and intrepret large volumes of data easily

eg: dashboard showing real time growth of stock-M

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4. What is data quality?

data quality refers to the accuracy Completeness and Consistency of data

eg: Customer db with accurate phonenos

5. What is data Remediation?

Data remediation is the process of Correcting errors in data

example: removing duplicate entries from a dataset.

6. What is dimension and measure?

Dimensions are Categorical fields used for grouping data

eg: region, product name

Measures are numerical field used for Calculations.

eg: Sales, profit

1) Tableau products

Tableau provide various products to support data vis-, analysis and sharing of Insights. The following are 5 imp Tableau products.

1. Tableau Desktop
2. Tableau Server
3. Tableau Online
4. Tableau public
5. Tableau prep

1. Tableau desktop

* TD is used to connect to diff data sources and create interactive visualaizations, reports and dashBoa

* It is mainly used by analyts and developers. Supports Data analysis with filters and Calculated fields

2. Tableau Server

* TS is a web product of Tableau that allow users to do Collaborative work on the same db at 2 diff places people can access it through mobile or desktop

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* Moreover it also provide security, user roles and permissions

3. Tableau Online

* Tableau online is a cloud based platform that perform the same functionality as Tableau Server.

* Moreover it provides same work of uploading the dataset on the server from anywhere via Internet conn.

4. Tableau public

* TP is a free product of Tableau which provides some limited functionalities for those who want to learn DV and understand Business intelligence

5. Tableau prep

* Tableau prep is a data preparation and ETL tool created by Tableau It is used to assist users with cleaning, shaping and combining data from large sources so that it is ready for analysis and visualization in TD.

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2. Tableau Workspace

Tableau Worvspace is a primary Interface where users connect to data, create visualaization and assemble dashboard and Stories.

1. Menu Bar,

* located at the top of windows
* It Contains options such as file, data, worvsheet, dashboard, story and help and is used to Save or export Tableau files.

2. Toolbar,

* provide quick access to Commonly used functions such as Save, undo, redo export and Swap r/c helping faster operations

3. Datapane,

* located at leftside of workspace
* It displays data Sources, tables, fields, dimensions and measures which can be dragged into worksheet

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4. Row and Column shelves,

* used to place fields for creating visualaization. They Control X and Y axis of the chart

5. Marks card,

* used to Control the appearance of data in the visualaization

* Include options like Color, size, label Tooltip and shape

6. Sheet tab,

* located at the bottom of the workspace. includes worksheet, dashboard and Story tabs which are used to navigate b/w diff sheets

7. show me panel,

* Suggest Suitable chart type based on Selected area

* It is especially usefull for beginners.

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chapter-2

1. Create, edit and remove hierarchy?

A Hierarchy in tableau is used to organaize related dimensions into levels. Therefore users can drill dow from summary data to detailed data

How to create hierarchy,

step1: open tableau

step2: In datapane, right click on a diamension (eg Country)

step3: click create hierarchy

step4: Give a name (eg: location

step5: Drag other fields into it. (State, City)

* We can also drag and drop the fields into tableau it will create hierarchy automatically

how to remove a hierarchy,

step 1: Right click on hierarchy

step 2: click remove hierarchy

* This does not delete data, It only removing the grouping

how to Edit a hierarchy,

1. Rename the hierarchy

step 1: In the datapane (leftside) right click on the hierarchy name

step 2: click rename and Type a new name

step 3: press enter

2. remove a field from hierarchy,

step 1: Expand the hierarchy
step 2: right click the field
step 3: click remove from hierarchy

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3. Add a field to existing hierarchy,

step1: find the field you want to add (eg: city)

step 2: Drag it into the hierarchy

step 3: drop it at the correct level.

(II) 4. Sets and Combined Sets

A set in tableau is a custom field that contains a subset of data based on condition or manual selection. It answers a Yes/No qs.

example: Customer who bought more than 5 times → IN the set → others OUT

Sets are dynamic and automatically update when the data changes, mainly used for comparative analysis, filter, highlting specific records

Steps, 1: click on datapane
2: right click on dimension
3. click create set
4: In dialouge box (Select values manually or put some conditions)

5. click ok

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Combined Set

* A Combined Set in Tableau : special type of set created by Combining two existing set that belong to same direction

* It uses logical operations such as AND, OR, NOT, EXCEPT to analyze the rela-ship b/w 2 diff subset of data

examples : * Customer who belong to both loyalty and high sales groups

* product that are popular but not profitable.

steps : 1: Create two individual set
2: right click on one set
3: create Combined set.
4: choose second set
5: select logical operator
6: click OK

Groups

* Grouping in tableau allow user to Combine related data values into a single group so that they can be analyzed and visualized together

* It is expecially usefull when there are many categories or when similar items needs to be treated as one.

Steps to Implement groups

step1 : open tableau tool and connect data

step 2 : drag and drop the one sheet of the connected dataset.

step3 : click on sheet1 to open tableau work sheet

step4 : In data pane right click dimension you want to group, select Create group

step 5 : Rename the group if needed click ok

step6 : Use the new group field in your worksheet

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1. Data preprocessing Technique

* Data prepro Tech is a essential step in DM that involves C, T, I, R to improving its quality and made for analysis.

* It Ensures Data Consis, accuracy is suite for analysis

1. Data cleaning

* It is the process of identifying and Correcting errors in a dataset It involves handling MV, removing duplicates, correcting outlier to ensure that data is more reliable.

Technique,
* filling MV
* Remove duplicate
* Smoothing noisy data

. eg: replacing a missing age value with the average age of all records

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2 Data Integration

Combine data from multiple sources into a single, consistent data.

Technique like: record linkage and data fusion help in combining data efficiently

Issues: Schema Integration, data redundancy data conflicts.

Example: Integrating customer data from different branches into one db

3. Data Transformation

* Converting the data into a format suitable for analysis. Techniques like normalization were used.
* The datas are converting from continuous to discrete

D, Normal → explain
D, aggregation →

eg: Transforming salary values into a specific range.

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4. Data reduction

* It reduces the dataset size while maintaining key information. This can be done through feature selection which chooses the most relevant features

dimensionality reduction : a technique that reduces the number of Variab

numerosity reduction : a technique that reduces the data point by sampling method

eg: selecting important attribute like name, age, income.

5. Data descretization

D.D Converts Continuos data into Categorical data. This step simplifies data analysis and helps algorithm to perform better on Categorical input.

Technique used : Bining method : group data-interval
histogram analysis : divid data base on frequency

clustering : groups

eg: Converting Continuos income values into Categories like L, M, high.